Dream, Symmetry, and Existence

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Abstract

A dream is not existence. A symmetry is not identity. An invariant is not a proof. Yet every proof begins with a hypothesis, every artwork begins with an image, and every observation begins by forgetting something. This note unifies four themes: symmetry, projection, hidden information, and creation.

1. Logic and Nonsense

Let

$$\mathcal{H} = \mathbb{C}^2$$

with orthonormal basis

$$|logic\rangle, \qquad |nonsense\rangle.$$

Define the involution

$$\Sigma|logic\rangle = |nonsense\rangle,$$
$$\Sigma|nonsense\rangle = |logic\rangle.$$

Then

$$\Sigma^2 = I, \qquad \Sigma^* = \Sigma.$$

The transformation does not prefer either side. Logic becomes nonsense. Nonsense becomes logic. Applying the map twice returns us to the beginning. The lesson is simple:

> A symmetry does not say that two things are identical. It says that the theory cannot distinguish them.

2. The Projection

Define

$$P = \frac{I + \Sigma}{2}.$$

Its invariant vector is

$$\psi_+ = \frac{|logic\rangle + |nonsense\rangle}{\sqrt{2}}.$$

For every

$$f = a|logic\rangle + b|nonsense\rangle,$$

we have

$$Pf = \frac{a+b}{\sqrt{2}}\psi_+.$$

The antisymmetric direction

$$\psi_- = \frac{|logic\rangle - |nonsense\rangle}{\sqrt{2}}$$

satisfies

$$P\psi_- = 0.$$

The disagreement has not been resolved. It has merely disappeared from observation. Projection preserves the common structure and hides the difference. What survives is the invariant. What vanishes enters the kernel.

3. Hidden Information

Let

$$Z = PZ + (I - P)Z.$$

The observed component is

$$Y = PZ.$$

The hidden component is

$$R = (I - P)Z.$$

Then

$$H(Z) = H(Y) + H(R|Y).$$

A constant output does not imply a constant reality. An observation may reveal nothing while a finite amount of information remains concealed. The projection says what is visible. The conditional entropy measures what is missing. The observer sees the image. The kernel remembers the distinction.

4. Different Phases

Consider

$$z_1 = e^{i\pi}, \qquad z_2 = e^{ie\ln\pi}, \qquad z_3 = e^{i\pi^2 e/2}.$$

Although their phases differ,

$$|z_1| = |z_2| = |z_3| = 1.$$

Different phases. One shared step. Different histories. One invariant geometry. The unit circle does not erase individuality. It reveals a property shared among distinct points.

5. The Dream

Suppose a system evolves inside a set K. Assume

$$x_0 \in K,$$

and

$$x_{t+1} \in G(w_t, x_t).$$

If every admissible action preserves K, then

$$x_t \in K \qquad \text{for all } t \geq 0.$$

The proof is induction. The mathematics is straightforward. The interpretation is not. One may assume a dream. One may even construct a model in which the dream remains admissible forever. But admissibility is not existence. Consistency is not reality. A hypothesis is not a world.

6. Creation

The mathematician seeks invariants. The dreamer seeks possibilities. The artist seeks form. All begin from the same act: selection. Every theorem projects. Every observation projects. Every artwork projects. Something is illuminated. Something remains hidden. Creation is therefore not the opposite of mathematics. Creation is another way of choosing what survives a projection.

7. Existence

The deepest mistake is to confuse these levels. A symmetry is not identity. A projection is not reality. An invariant is not existence. And yet none of those are meaningless. Without symmetry there is no structure. Without projection there is no observation. Without hypothesis there is no proof. Without imagination there is nothing to prove.

Conclusion

The journey may be written as

$$\text{Symmetry} \longrightarrow \text{Invariance} \longrightarrow \text{Observation} \longrightarrow \text{Existence}.$$

Logic and nonsense exchange places. Projection preserves what they share. Entropy measures what remains hidden. The dream proposes a world. Existence must still be demonstrated.

I dare to dream. I can assume the dream. I still have to prove existence.